Completion of the proof of Lemma 1: tamely ramified sheaves extend over the boundary of Takeuchi's compactification

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Goal. This note completes the proof of Theorem 2 ("Theorem 2"). The two-case argument of the thesis reduces everything, in the ramified case $\mathfrak{m} > 0$, to the following extension lemma, whose proof in the thesis is currently a placeholder (`ProofOfReducedTheorem.tex`): a locally constant sheaf $\mathcal{F}^{[d]}$ on $U^{(d)}$ which is *tamely* ramified along the boundary divisor $H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)}$ extends to a locally constant sheaf on all of $\tilde{C}_{\mathfrak{m}}^{(d)}$.

The proof below proceeds in two independent steps: by Zariski–Nagata purity, it suffices to show the corresponding torsor is *unramified* at the generic point η_0 of H (Proposition 3); then tameness at η_0 is upgraded to unramifiedness (Proposition 4) — the wild part is killed by tameness alone, and the tame part by a Kummer-theoretic valuation computation using the projective bundle structure of $C^{(d)}$ over Pic_C^d.

All numbered references to results *not* proved in this note (Takeuchi's compactification, the degree condition, purity inputs, etc.) carry the numbering of the compiled thesis (`main.pdf`).

We keep the notation of Section 4.4: C is a projective smooth geometrically connected curve over the perfect field k, $\mathfrak{m} = \sum n_i P_i > 0$ is a modulus with complement $U = C \setminus \mathfrak{m}$, and d is an integer satisfying (10). We write $Z_0 = Z_0(\mathfrak{m}, d) \subset C^{(d)}$ for the center of the blowup $\pi : X_{\mathfrak{m},d} \to C^{(d)}$, E_0 for its exceptional divisor with generic point η_0, $\tilde{C}_{\mathfrak{m}}^{(d)} \subseteq X_{\mathfrak{m},d}$ for Takeuchi's compactification (Theorem 80), and $H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)} = E_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)}$ for the boundary divisor. Throughout, p denotes the characteristic of k (the case $p = 0$ only simplifies the arguments below).

Lemma 1. *If $\mathcal{F}^{[d]}$ is a locally constant sheaf on $U^{(d)}$ which is tamely ramified along the boundary divisor $H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)}$, then $\mathcal{F}^{[d]}$ extends to a locally constant sheaf $\tilde{\mathcal{F}}^{[d]}$ on $\tilde{C}_{\mathfrak{m}}^{(d)}$.*

The proof occupies the rest of this subsection and proceeds in two independent steps. First (Proposition 3), by the Zariski–Nagata purity of the branch locus, extending $\mathcal{F}^{[d]}$ across H is possible as soon as the corresponding torsor is *unramified* at the generic point η_0 of H; a priori tameness is weaker than this. Second (Proposition 4), we show that at η_0 tameness in fact forces unramifiedness. This is where the global geometry enters: the wild (p-primary) part of the torsor is killed at η_0 by tameness alone, while the tame (prime-to-p) part is killed by an explicit Kummer-theoretic computation of valuations, using the projective bundle structure of $C^{(d)}$ over Pic_C^d. Note that the second step is exactly where "tame" is upgraded to "unramified": for a *sub*modulus $\mathfrak{n} \subsetneq \mathfrak{m}$ the analogous statement at $\eta_{\mathfrak{n}}$ is false in general (the computations of Section 3 produce genuinely, tamely, ramified torsors there), and the argument below makes visible why the full modulus and the completed linear systems are needed.

We first record the relevant geometry of the pair $(\tilde{C}^{(d)}_{\mathfrak{m}}, H)$.

Lemma 2. *With notation as above:*

1. *$\tilde{C}^{(d)}_{\mathfrak{m}}$ is an integral scheme, smooth over k.*
2. *H is a nonempty open subscheme of E_0. In particular H is irreducible of codimension 1 in $\tilde{C}^{(d)}_{\mathfrak{m}}$, its generic point is η_0, and $\eta_0 \in \tilde{C}^{(d)}_{\mathfrak{m}}$.*
3. *η_0 is the unique point of $\tilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$ whose local ring has dimension 1, and $\mathcal{O}_{\tilde{C}^{(d)}_{\mathfrak{m}},\eta_0} = \mathcal{O}_{X_{\mathfrak{m},d},\eta_0}$ is a discrete valuation ring with fraction field $K := k(C^{(d)})$.*

Proof. (1) $\tilde{C}^{(d)}_{\mathfrak{m}}$ is an open subscheme of $X_{\mathfrak{m},d}$, which is integral as the blowup of the integral scheme $C^{(d)}$; hence $\tilde{C}^{(d)}_{\mathfrak{m}}$ is integral. By Theorem 80, $\tilde{C}^{(d)}_{\mathfrak{m}}$ is a projective space bundle over $\mathrm{Pic}^d_{C,\mathfrak{m}}$. The latter is smooth over k: $\mathrm{Pic}^0_{C,\mathfrak{m}}$ is a smooth group scheme and $\mathrm{Pic}^d_{C,\mathfrak{m}}$ is a torsor under it (see the properties of the generalized Picard scheme recalled above). A projective space bundle over a smooth k-scheme is smooth over k.

(2) Since $\tilde{C}^{(d)}_{\mathfrak{m}} \hookrightarrow X_{\mathfrak{m},d}$ is an open immersion, its base change $H = E_0 \times_{C^{(d)}} \tilde{C}^{(d)}_{\mathfrak{m}} = E_0 \times_{X_{\mathfrak{m},d}} \tilde{C}^{(d)}_{\mathfrak{m}} \hookrightarrow E_0$ is an open immersion as well. To see $H \neq \emptyset$, suppose otherwise; then $U^{(d)} = \tilde{C}^{(d)}_{\mathfrak{m}}$ and $\Phi_d = \tilde{\Phi}_d$ would be a projective space bundle. But the fibers of Φ_d are affine spaces of dimension $d - \deg \mathfrak{m} - g + 1 \geq 1$ (Section 4.3; positivity follows from (10)), and a positive dimensional affine space over a field is not proper — a contradiction. A nonempty open subscheme of the irreducible scheme E_0 is irreducible, dense, and contains the generic point η_0 of E_0.

(3) By Theorem 80, $\tilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)} = H$, which by (2) is irreducible of codimension 1 with generic point η_0; every other point of H is a proper specialization of η_0 and hence has local ring of dimension ≥ 2. By (1), $\mathcal{O}_{\tilde{C}^{(d)}_{\mathfrak{m}},\eta_0}$ is a regular local ring of dimension 1, i.e. a discrete valuation ring, and it equals $\mathcal{O}_{X_{\mathfrak{m},d},\eta_0}$ because $\tilde{C}^{(d)}_{\mathfrak{m}}$ is open in $X_{\mathfrak{m},d}$. Its fraction field is the function field of $X_{\mathfrak{m},d}$, which equals $K = k(C^{(d)})$ since π is birational. □

We denote by v_{η_0} the discrete valuation on K associated to $\mathcal{O}_{\tilde{C}^{(d)}_{\mathfrak{m}},\eta_0}$, and by $\widehat{K}_{\eta_0}$ the fraction field of the completion.

Step 1: Purity, and extending unramified torsors

Proposition 3. *Let G be a finite abelian group, let $\rho : \pi_1^{et}(U^{(d)}, \bar{u}) \to G$ be a continuous homomorphism and let $\mathcal{Q}$ be the corresponding G-torsor on $U^{(d)}_{\acute{e}t}$ (Corollary 14). If $\mathcal{Q}$ is unramified over $\tilde{C}^{(d)}_{\mathfrak{m}}$ at (H, η_0) in the sense of Definition 54, then ρ factors, necessarily uniquely, through the canonical surjection*

$$\iota_* : \pi_1^{et}(U^{(d)}, \bar{u}) \twoheadrightarrow \pi_1^{et}(\tilde{C}^{(d)}_{\mathfrak{m}}, \bar{u}),$$

where $\iota : U^{(d)} \hookrightarrow \tilde{C}^{(d)}_{\mathfrak{m}}$ is the open immersion. Consequently, $\mathcal{Q}$ extends to a G-torsor on $(\tilde{C}^{(d)}_{\mathfrak{m}})_{\acute{e}t}$, uniquely up to isomorphism.

Proof. **Surjectivity of ι_*.** We first note that for every connected finite étale cover $T \to \tilde{C}_{\mathfrak{m}}^{(d)}$, the restriction $T \times_{\tilde{C}_{\mathfrak{m}}^{(d)}} U^{(d)}$ is connected: T is étale over the regular scheme $\tilde{C}_{\mathfrak{m}}^{(d)}$, hence regular, and being connected it is irreducible; $T \times_{\tilde{C}_{\mathfrak{m}}^{(d)}} U^{(d)}$ is a nonempty open subscheme of T (nonempty since $T \to \tilde{C}_{\mathfrak{m}}^{(d)}$ is surjective and $U^{(d)}$ is dense), hence irreducible, hence connected. This implies surjectivity of ι_*: if the image were a proper closed subgroup, it would be contained in a proper open subgroup $M \leq \pi_1^{et}(\tilde{C}_{\mathfrak{m}}^{(d)}, \bar{u})$, and the connected cover corresponding to M would pull back to a disconnected cover of $U^{(d)}$ (the image of $\pi_1^{et}(U^{(d)})$ does not act transitively on $\pi_1^{et}(\tilde{C}_{\mathfrak{m}}^{(d)})/M$), a contradiction.

The connected cover and its normalization. Let $G' = \rho\Big(\pi_1^{et}(U^{(d)}, \bar{u})\Big) \subseteq G$ and let $V \to U^{(d)}$ be the connected finite étale Galois cover with group G' corresponding to the open normal subgroup $\ker \rho$ (Theorem 13). As a $U^{(d)}$-scheme, $\mathcal{Q}$ is isomorphic to a disjoint union of $[G : G']$ copies of V; hence at the generic point η_0 the finite separable $\widehat{K}_{\eta_0}$-algebras of $\mathcal{Q}$ and of V have the same field factors, and V is unramified over $\tilde{C}_{\mathfrak{m}}^{(d)}$ at (H, η_0).

V is étale over the smooth scheme $U^{(d)}$, hence smooth over k; being connected it is integral, and $K(V)/K$ is a finite separable field extension. Let $\nu : W \to \tilde{C}_{\mathfrak{m}}^{(d)}$ be the normalization of $\tilde{C}_{\mathfrak{m}}^{(d)}$ in $K(V)$, i.e. the relative spectrum of the integral closure of $\mathcal{O}_{\tilde{C}_{\mathfrak{m}}^{(d)}}$ in $K(V)$. Since $\tilde{C}_{\mathfrak{m}}^{(d)}$ is of finite type over a field it is Nagata ([Stacks, Tag 0335]), so ν is a finite morphism, and W is integral and normal. Over the open $U^{(d)}$, the scheme V is finite over $U^{(d)}$, normal, with function field $K(V)$; by uniqueness of normalization, $\nu^{-1}(U^{(d)}) \cong V$ as $U^{(d)}$-schemes.

ν is étale, by purity of the branch locus. We apply [Stacks, Tag 0BMB] at each point $w \in W$ with image $y = \nu(w)$. The hypotheses hold: $\mathcal{O}_{W,w}$ is normal; $\mathcal{O}_{\tilde{C}_{\mathfrak{m}}^{(d)},y}$ is regular (Lemma 2); ν is finite, hence quasi-finite at w; and $\dim(\mathcal{O}_{W,w}) = \dim(\mathcal{O}_{\tilde{C}_{\mathfrak{m}}^{(d)},y})$, since for a finite dominant morphism of integral schemes of finite type over a field one has $\dim(\mathcal{O}_{W,w}) = \dim W - \dim \overline{\{w\}}$, $\dim W = \dim \tilde{C}_{\mathfrak{m}}^{(d)}$ and $\dim \overline{\{w\}} = \dim \overline{\{y\}}$. It remains to check the codimension-one condition: let $w' \rightsquigarrow w$ be a specialization with $\dim(\mathcal{O}_{W,w'}) = 1$; then $y' = \nu(w')$ satisfies $\dim(\mathcal{O}_{\tilde{C}_{\mathfrak{m}}^{(d)},y'}) = 1$ by the same dimension count. If $y' \in U^{(d)}$, then ν is étale at w' since $\nu^{-1}(U^{(d)}) = V$ is étale over $U^{(d)}$. Otherwise $y' \in \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)}$, and by Lemma 2(3) necessarily $y' = \eta_0$; then $\mathcal{O}_{W,w'}$ is a localization of the integral closure of the discrete valuation ring $\mathcal{O}_{\tilde{C}_{\mathfrak{m}}^{(d)},\eta_0}$ in $K(V)$ at one of its maximal ideals, and the hypothesis that V is unramified at (H, η_0) says precisely (Definition 27) that the ramification index there is 1 and the residue extension is separable, i.e. that ν is unramified at w'. If $\dim(\mathcal{O}_{W,w}) = 0$ then w is the generic point of W and ν is étale at w because $K(V)/K$ is finite separable. In all cases [Stacks, Tag 0BMB] applies, and we conclude that $\nu : W \to \tilde{C}_{\mathfrak{m}}^{(d)}$ is finite étale.

Factorization. The geometric point $\bar{u}$ of $U^{(d)}$ is also a geometric point of $\tilde{C}_{\mathfrak{m}}^{(d)}$, and $W_{\bar{u}} = V_{\bar{u}}$ as finite sets. The action of $\pi_1^{et}(U^{(d)}, \bar{u})$ on $V_{\bar{u}}$ is the restriction along ι_* of the action of $\pi_1^{et}(\tilde{C}_{\mathfrak{m}}^{(d)}, \bar{u})$ on $W_{\bar{u}}$, because $V = W \times_{\tilde{C}_{\mathfrak{m}}^{(d)}} U^{(d)}$. Since V is Galois with group G', the kernel of the first action is exactly $\ker \rho$; hence $\ker(\iota_*) \subseteq \ker \rho$. As ι_* is surjective, ρ factors as $\rho = \tilde{\rho} \circ \iota_*$ for a unique continuous $\tilde{\rho} : \pi_1^{et}(\tilde{C}_{\mathfrak{m}}^{(d)}, \bar{u}) \to G' \subseteq G$. Let $\tilde{\mathcal{Q}}$ be the G-torsor on $(\tilde{C}_{\mathfrak{m}}^{(d)})_{\text{ét}}$ corresponding to $\tilde{\rho}$ (Corollary 14); then $\iota^* \tilde{\mathcal{Q}} \cong \mathcal{Q}$ since the corresponding characters of $\pi_1^{et}(U^{(d)}, \bar{u})$ agree. Finally, if $\tilde{\mathcal{Q}}_1, \tilde{\mathcal{Q}}_2$ are two extensions of $\mathcal{Q}$, their characters $\tilde{\rho}_1, \tilde{\rho}_2$ satisfy $\tilde{\rho}_1 \circ \iota_* = \tilde{\rho}_2 \circ \iota_*$, hence are equal by surjectivity of ι_*, and $\tilde{\mathcal{Q}}_1 \cong \tilde{\mathcal{Q}}_2$. □

Step 2: At the boundary, tame implies unramified

Proposition 4. *Let G be a finite abelian group and let $\mathcal{Q}$ be a G-torsor on $U^{(d)}_{ét}$. If $\mathcal{Q}$ is tamely ramified over $\tilde{C}^{(d)}_{\mathfrak{m}}$ at (H, η_0), then $\mathcal{Q}$ is unramified at (H, η_0).*

Note that no bound on the ramification of $\mathcal{Q}$ along $\mathfrak{m}$ is assumed; being a torsor on all of $U^{(d)}$ is enough.

Proof. As in Section 3 (see the discussion following Definition 55), the restriction of $\mathcal{Q}$ to $\mathrm{Spec}(\widehat{K}_{\eta_0})$ decomposes as a disjoint union of copies of $\mathrm{Spec}(F)$ for a finite Galois extension $F/\widehat{K}_{\eta_0}$ with Galois group a subgroup of G, and corresponds to a continuous homomorphism

$$\rho : G_{\widehat{K}_{\eta_0}} \to G$$

with image $\mathrm{Gal}(F/\widehat{K}_{\eta_0})$. Let $I \subseteq G_{\widehat{K}_{\eta_0}}$ be the inertia subgroup. Then $\mathcal{Q}$ is unramified at (H, η_0) if and only if $\rho(I) = \{1\}$, and $\rho(I)$ equals the inertia subgroup G_0 of $\mathrm{Gal}(F/\widehat{K}_{\eta_0})$.

Reduction to k algebraically closed. Let k'/k be a finite separable extension. The projection $\mathrm{pr} : (\tilde{C}^{(d)}_{\mathfrak{m}})_{k'} \to \tilde{C}^{(d)}_{\mathfrak{m}}$ is (finite) étale, and it identifies $(\tilde{C}^{(d)}_{\mathfrak{m}})_{k'} \setminus (U^{(d)})_{k'}$ with $\mathrm{pr}^{-1}(H)$. Since $C^{(n)}$ is geometrically integral for every n (see the Preliminaries), so are Z_0, $C^{(d)}$ and hence E_0 (the exceptional divisor of a blowup of a geometrically integral smooth scheme along a geometrically integral smooth center); therefore $\mathrm{pr}^{-1}(H)$ is irreducible with a unique generic point η' lying over η_0, and pr is étale at η'. By Lemma 59 (and its proof: the completed local extensions at η' and at η_0 differ by the unramified base change $\widehat{K}_{\eta'}/\widehat{K}_{\eta_0}$), the torsor $\mathrm{pr}^{-1}\mathcal{Q}$ is tamely ramified, resp. unramified, at η' if and only if $\mathcal{Q}$ is tamely ramified, resp. unramified, at η_0. Passing to the colimit over all finite separable k'/k (recall k is perfect, so $\bar{k} = k^{sep}$), we may and do assume for the rest of the proof that k is algebraically closed. In particular the points of $\mathfrak{m}$ are k-rational; write $\mathrm{supp}(\mathfrak{m}) = \{P_1, \dots, P_s\}$.

Splitting off the wild part. By the structure theorem for finite abelian groups, write $G = G_p \times G'$ with G_p a p-group and $|G'|$ prime to p (if $p = 0$, take $G_p = \{1\}$). Then $\rho(I) = \{1\}$ if and only if both compositions $\rho_p := \mathrm{pr}_{G_p} \circ \rho$ and $\rho' := \mathrm{pr}_{G'} \circ \rho$ kill I.

Since $\mathcal{Q}$ is tamely ramified at (H, η_0), the extension $F/\widehat{K}_{\eta_0}$ is tamely ramified with respect to $\mathcal{O}_{\tilde{C}^{(d)}_{\mathfrak{m}}, \eta_0}$ (Definition 27): its residue extension is separable and its ramification index e is prime to p. For a Galois extension of complete discrete valuation rings with separable residue extension, the inertia group G_0 has order equal to the ramification index ([Ser79], Ch. IV); hence $\rho(I) = G_0$ is a group of order prime to p. Its projection to the p-group G_p is then trivial, that is:

$$\rho_p(I) = \{1\}, \qquad \rho(I) \subseteq \{1\} \times G'.$$

It therefore remains to prove $\rho'(I) = \{1\}$.

The prime-to-p part, via Kummer theory. Let $N = |G'|$, which is invertible in $k = \bar{k}$, so $\mu_N \subset k$. Since G' is abelian, $\rho'(I) = \{1\}$ holds if and only if $\chi(\rho'(I)) = \{1\}$ for every character $\chi : G' \to \mu_N(k)$. Fix such a χ and let $e \mid N$ be the order of the character

$$\chi \circ \rho'_{glob} : \pi_1^{et}(U^{(d)}, \bar{u}) \longrightarrow \mu_e(k),$$

where ρ'_{glob} denotes the global character of $\mathcal{Q}/G_p$ (Proposition 15) whose local restriction at η_0 is ρ'. It suffices to show that the μ_e-torsor corresponding to $\chi \circ \rho'_{glob}$ is unramified at (H, η_0); indeed, the inertia I then acts trivially on it, i.e. $\chi(\rho'(I)) = \{1\}$.

Restrict $\chi \circ \rho'_{glob}$ to the absolute Galois group of the function field $K = k(U^{(d)}) = k(C^{(d)})$. Since $\mu_e \subset K$, Kummer theory identifies $H^1(K, \mu_e) \cong K^\times/(K^\times)^e$; let $F \in K^\times$ be a representative of the resulting class, so that the corresponding cyclic cover of $\mathrm{Spec}(K)$ is $\mathrm{Spec}\big(K[y]/(y^e - F)\big)$, and F is well defined up to e-th powers. Accordingly, $\mathrm{div}_{C^{(d)}}(F)$ is well defined modulo $e \cdot \mathrm{Div}(C^{(d)})$. We will show:

$$v_{\eta_0}(F) \equiv 0 \pmod{e}. \tag{1}$$

Let us first explain why (1) finishes the proof. Let ϖ be a uniformizer of $B_0 := \mathcal{O}_{\tilde{C}^{(d)}_{\mathfrak{m}}, \eta_0}$ and write $v_{\eta_0}(F) = ea$; replacing F by $F\varpi^{-ea}$ (an allowed change modulo nothing — it changes neither the class in $H^1(\widehat{K}_{\eta_0}, \mu_e)$ nor the extension, since it changes F by the e-th power $\big(\varpi^{-a}\big)^e$), we may assume $F \in \widehat{B}_0^\times$, where $\widehat{B}_0$ is the completion. Then

$$B := \widehat{B}_0[y]/(y^e - F)$$

is finite étale over $\widehat{B}_0$: it is finite free, and the derivative ey^{e-1} is a unit in B because e is invertible in k and $y^e = F$ is a unit. Consequently $\widehat{K}_{\eta_0}[y]/(y^e - F) = B \otimes_{\widehat{B}_0} \widehat{K}_{\eta_0}$ is a product of finite separable field extensions of $\widehat{K}_{\eta_0}$, each unramified with respect to $\widehat{B}_0$ (Definition 27; compare Theorem 30(1),(2), whose proof for e prime to p does not use perfectness of the residue field). Thus the μ_e-torsor of $\chi \circ \rho'_{glob}$ is unramified at (H, η_0), as desired.

The rest of the proof establishes (1), in three claims. For each $1 \le i \le s$ let

$$\Delta_{P_i} := Z_0(P_i, d) \subset C^{(d)}$$

be the prime divisor obtained as the (closed) image of the morphism $C^{(d-1)} \to C^{(d)}$, $D \mapsto D + P_i$; thus $C^{(d)} \setminus U^{(d)} = \bigcup_{i=1}^s \Delta_{P_i}$, because a divisor of degree d fails to be prime to $\mathfrak{m}$ exactly when it contains some P_i. Set

$$c_i := v_{\Delta_{P_i}}(F) \in \mathbb{Z}.$$

Claim 5. $\mathrm{div}_{C^{(d)}}(F) \equiv \sum_{i=1}^s c_i \, \Delta_{P_i} \pmod{e \cdot \mathrm{Div}(C^{(d)})}$.

Proof. Let $W \subset C^{(d)}$ be a prime divisor different from all the Δ_{P_i}, with generic point ξ_W. Then $\xi_W \in U^{(d)}$, and the μ_e-torsor of $\chi \circ \rho'_{glob}$, being a torsor on all of $U^{(d)}$, is finite étale over a neighborhood of ξ_W, hence unramified at (W, ξ_W) (Definition 54, with ambient scheme $C^{(d)}$). On the other hand, if $e \nmid v_W(F)$, then in any extension of the valuation v_W to $K(y)/(y^e - F)$ one has $v(y) = v_W(F)/e \notin v_W(K^\times)$, so the ramification index at (W, ξ_W) is $e/\gcd(e, v_W(F)) > 1$ — contradicting unramifiedness. Hence $e \mid v_W(F)$ for every such W. □

Let ζ denote the generic point of Z_0 and let $A := \mathcal{O}_{C^{(d)},\zeta}$, a regular local ring of dimension $\deg \mathfrak{m}$ (the codimension of $Z_0 \cong C^{(d-\deg \mathfrak{m})}$ in $C^{(d)}$) with maximal ideal $\mathfrak{m}_\zeta = I_{Z_0,\zeta}$ and residue field $\kappa(\zeta)$. Since E_0 dominates Z_0, the center of v_{η_0} on $C^{(d)}$ is ζ, i.e. $A \subseteq \mathcal{O}_{X_{\mathfrak{m},d},\eta_0}$ with $\mathfrak{m}_\zeta = A \cap \mathfrak{m}_{\eta_0}$; in particular $v_{\eta_0} \ge 0$ on A and $v_{\eta_0} = 0$ on $A^\times$. For $0 \ne g \in A$ let $\mathrm{ord}_\zeta(g) := \max\{n : g \in \mathfrak{m}_\zeta^n\}$ be the order of vanishing of g along Z_0.

Claim 6. *For every $0 \ne g \in A$ we have $v_{\eta_0}(g) = \mathrm{ord}_\zeta(g)$.*

Proof. This is the general form of the local computation carried out explicitly (in coordinates) in Section 3; we recall the argument. Since $\mathrm{Spec}(A) \to C^{(d)}$ is flat and $I_{Z_0} \cdot A = \mathfrak{m}_\zeta$, blowups commute with this base change ([Stacks, Tag 0805]):

$$X_{\mathfrak{m},d} \times_{C^{(d)}} \mathrm{Spec}(A) \cong \mathrm{Bl}_{\mathfrak{m}_\zeta}(\mathrm{Spec}\, A),$$

the exceptional divisor pulls back to the exceptional divisor $E_A = \mathbb{P}^{\deg \mathfrak{m}-1}_{\kappa(\zeta)}$, whose generic point maps to η_0 with the same local ring. So we may compute v_{η_0} on the blowup of the regular local ring A at its maximal ideal. Choose a regular system of parameters $u_1, \ldots, u_n$ of A ($n = \deg \mathfrak{m}$) and consider the chart $A' = A[\mathfrak{m}_\zeta/u_n]$. One has $\mathfrak{m}_\zeta A' = (u_n)$ and, since A is regular, $A'/(u_n) \cong \kappa(\zeta)[\bar{v}_1, \ldots, \bar{v}_{n-1}]$ is a polynomial ring in the classes of $v_j := u_j/u_n$; in particular $(u_n) \subset A'$ is prime, the localization $A'_{(u_n)} = \mathcal{O}_{X_{\mathfrak{m},d},\eta_0}$ is a discrete valuation ring with uniformizer u_n, and E_A meets this chart in its generic point. Now let $r = \mathrm{ord}_\zeta(g)$ and write $g = G_r(u_1, \ldots, u_n) + g'$ where G_r is a homogeneous form of degree r with unit or zero coefficients, nonzero as an element of $\mathrm{gr}^r_{\mathfrak{m}_\zeta}(A) \cong \kappa(\zeta)[\bar{u}_1, \ldots, \bar{u}_n]_r$, and $g' \in \mathfrak{m}_\zeta^{r+1}$. In the chart,

$$g = u_n^r\big(G_r(v_1, \ldots, v_{n-1}, 1) + u_n h\big), \qquad h \in A',$$

and the residue of $G_r(v_1, \ldots, v_{n-1}, 1)$ in $A'/(u_n)$ is the dehomogenization of the nonzero form G_r, hence nonzero; therefore $G_r(v, 1) + u_n h$ is a unit in $A'_{(u_n)}$ and $v_{\eta_0}(g) = r$. □

Claim 7. *For each i, every local equation $w_i \in A$ of Δ_{P_i} at ζ satisfies $\mathrm{ord}_\zeta(w_i) = 1$; consequently $v_{\eta_0}(w_i) = 1$.*

Proof. First note $\zeta \in \Delta_{P_i}$: every divisor $D \geq \mathfrak{m}$ contains P_i, i.e. $Z_0 \subseteq \Delta_{P_i}$. Since A is regular local, hence a unique factorization domain, the height one prime of Δ_{P_i} at ζ is principal; fix a generator w_i.

By Corollary 82, the addition morphism

$$\sigma : C^{(\deg \mathfrak{m})} \times_k C^{(d-\deg \mathfrak{m})} \longrightarrow C^{(d)}$$

is étale at the point $\zeta' := (z_{\mathfrak{m}}, \xi')$, where $z_{\mathfrak{m}} \in C^{(\deg \mathfrak{m})}$ is the closed point corresponding to $\mathfrak{m}$ and ξ' is the generic point of $C^{(d-\deg \mathfrak{m})}$; moreover $\sigma(\zeta') = \zeta$, since σ restricted to $\{z_{\mathfrak{m}}\} \times C^{(d-\deg \mathfrak{m})}$ is the isomorphism onto Z_0 defining it. Because σ is étale at ζ', the local homomorphism $A \to A'' := \mathcal{O}_{C^{(\deg \mathfrak{m})} \times C^{(d-\deg \mathfrak{m})}, \zeta'}$ is faithfully flat with $\mathfrak{m}_\zeta A'' = \mathfrak{m}_{\zeta'}$, so $\mathrm{ord}_\zeta(g) = \mathrm{ord}_{\zeta'}(\sigma^* g)$ for all $g \in A$.

We identify $\sigma^* w_i$ near ζ'. Set-theoretically,

$$\sigma^{-1}(\Delta_{P_i}) = \Big(\Delta'_{P_i} \times_k C^{(d-\deg \mathfrak{m})}\Big) \cup \Big(C^{(\deg \mathfrak{m})} \times_k \Delta''_{P_i}\Big),$$

where $\Delta'_{P_i} \subset C^{(\deg \mathfrak{m})}$ and $\Delta''_{P_i} \subset C^{(d-\deg \mathfrak{m})}$ are the analogous divisors of divisors containing P_i. The second component does not pass through ζ' (the generic divisor of degree $d - \deg \mathfrak{m}$ does not contain P_i), so near ζ' the scheme-theoretic preimage $\sigma^{-1}(\Delta_{P_i}) = V(\sigma^* w_i)$ is supported on the prime divisor $\Delta'_{P_i} \times_k C^{(d-\deg \mathfrak{m})}$ (a product of integral schemes, integral by Theorem 47). Moreover $\sigma^{-1}(\Delta_{P_i})$ is reduced, being the étale pullback (near ζ') of the reduced scheme Δ_{P_i}. Hence $\sigma^* w_i$ is, up to a unit at ζ', a local equation $w'_i \otimes 1$ of $\Delta'_{P_i} \times_k C^{(d-\deg \mathfrak{m})}$, where w'_i is a local equation of Δ'_{P_i} at $z_{\mathfrak{m}}$. Writing $B := \mathcal{O}_{C^{(\deg \mathfrak{m})}, z_{\mathfrak{m}}}$ and noting that A'' is a localization of $B \otimes_k \kappa(\xi')$ at the prime $\mathfrak{m}_B \otimes \kappa(\xi')$, the homomorphism $B \to A''$ is flat with $\mathfrak{m}_B A'' = \mathfrak{m}_{\zeta'}$, so

$$\mathrm{ord}_\zeta(w_i) = \mathrm{ord}_{\zeta'}(w'_i \otimes 1) = \mathrm{ord}_{z_{\mathfrak{m}}}(w'_i),$$

and we are reduced to showing that $w'_i \notin \mathfrak{m}^2_{z_{\mathfrak{m}}}$.

This is a computation at the closed point $z_{\mathfrak{m}} = \sum_j n_j P_j$ of $C^{(\deg \mathfrak{m})}$, for which we use the local description of symmetric powers from the Preliminaries (proof of the smoothness of $C^{(n)}$): choosing pairwise disjoint opens $U_j \ni P_j$ and local coordinates t_j on U_j vanishing exactly at P_j, a Zariski-open neighborhood of $z_{\mathfrak{m}}$ decomposes as $\prod_j U_j^{(n_j)}$, and the completed local ring is

$$\widehat{\mathcal{O}}_{C^{(\deg \mathfrak{m})},z_{\mathfrak{m}}} \cong k[[\,\varepsilon_l^{(j)}\ :\ 1 \le j \le s,\ 1 \le l \le n_j\,]],$$

where $\varepsilon_l^{(j)}$ is the l-th elementary symmetric function of the coordinates $t_j(x_1), \dots, t_j(x_{n_j})$ of the points of the j-th block. Under the product decomposition, Δ'_{P_i} is the preimage of the divisor $\{D_i : D_i \ni P_i\} \subset U_i^{(n_i)}$, and the invariant function

$$N_i := \prod_{l=1}^{n_i} t_i(x_l) = \varepsilon_{n_i}^{(i)}$$

is a regular function near $z_{\mathfrak{m}}$ vanishing exactly on Δ'_{P_i} (as t_i vanishes only at P_i on U_i). In particular w'_i divides $\varepsilon_{n_i}^{(i)}$ in $\mathcal{O}_{C^{(\deg \mathfrak{m})},z_{\mathfrak{m}}}$. Orders with respect to the maximal ideal are computed in the completion and are additive (the associated graded ring is a polynomial ring, hence a domain); since $\varepsilon_{n_i}^{(i)}$ is one of the coordinates of the power series ring, its order is 1, and therefore $\operatorname{ord}_{z_{\mathfrak{m}}}(w'_i) = 1$ as well. The last statement of the claim follows from Claim 6. $\square$

Claim 8. $\sum_{i=1}^{s} c_i \equiv 0 \pmod{e}$.

Proof. Here we use the Abel–Jacobi morphism *with trivial modulus*

$$\varphi_d : C^{(d)} \longrightarrow \operatorname{Pic}^d_C .$$

Since $d \ge 2g - 1 + \deg \mathfrak{m} \ge 2g$ and k is algebraically closed (so C has a rational point), $C^{(d)}$ is a projective space bundle over Pic^d_C and the fiber over a point $[\mathcal{L}]$ is the complete linear system $|\mathcal{L}| = \mathbb{P}\big(H^0(C, \mathcal{L})\big)$ (Section 4.3). Let ξ be the generic point of Pic^d_C, with residue field $\kappa(\xi)$, and let

$$P_\xi := \varphi_d^{-1}(\xi) \cong \mathbb{P}^{d-g}_{\kappa(\xi)}$$

be the generic fiber; it is a localization of $C^{(d)}$, so for every point $w \in P_\xi$ we have $\mathcal{O}_{P_\xi,w} = \mathcal{O}_{C^{(d)},w}$. In particular the codimension-one points of P_ξ are exactly the generic points of the prime divisors of $C^{(d)}$ that dominate Pic^d_C, and for such a divisor W with generic-fiber components $w \in W \cap P_\xi$,

$$\operatorname{div}_{P_\xi}\Big(F|_{P_\xi}\Big) = \sum_W v_W(F) \cdot \big[W \cap P_\xi\big], \tag{2}$$

where the sum is over the prime divisors of $C^{(d)}$ dominating Pic^d_C and $[W \cap P_\xi]$ denotes the corresponding reduced cycle of codimension one on P_ξ. (The generic point of $C^{(d)}$ lies in P_ξ, so F restricts to a nonzero rational function on P_ξ.)

Each Δ_{P_i} dominates Pic^d_C: for every $\mathcal{L}$ of degree d one has $h^0(\mathcal{L}(-P_i)) = d - g \ge 1$, so every fiber $|\mathcal{L}|$ contains divisors through P_i. Moreover, inside the generic fiber,

$$\Delta_{P_i} \cap P_\xi = \big\{D \in |\mathcal{L}_\xi|\ :\ D \ge P_i\big\} = \mathbb{P}\Big(H^0(C_{\kappa(\xi)}, \mathcal{L}_\xi(-P_i))\Big) \subset \mathbb{P}\Big(H^0(C_{\kappa(\xi)}, \mathcal{L}_\xi)\Big),$$

a linear subspace of codimension one (as $\deg \mathcal{L}_\xi(-P_i) = d - 1 \ge 2g - 1$, so $h^0(\mathcal{L}_\xi(-P_i)) = d - g = h^0(\mathcal{L}_\xi) - 1$). In particular $\Delta_{P_i} \cap P_\xi$ is irreducible and, as a reduced linear subvariety of projective space, it has degree 1.

Now take degrees in (2). A principal divisor on $\mathbb{P}^{d-g}_{\kappa(\xi)}$ has degree 0 (its homogeneous numerator and denominator have equal degrees). By Claim 5, every dominating prime divisor W other than the Δ_{P_i} contributes a multiple of e, while Δ_{P_i} contributes $c_i \cdot \deg[\Delta_{P_i} \cap P_\xi] = c_i$. Hence

$$0 = \deg \operatorname{div}_{P_\xi}\left(F|_{P_\xi}\right) \equiv \sum_{i=1}^{s} c_i \pmod{e}. \qquad \square$$

We can now conclude. Since A is a unique factorization domain, we may write

$$F = u \cdot \prod_j w_j^{a_j}, \qquad u \in A^\times,$$

where w_j runs over generators of the height-one primes of A, i.e. over local equations at ζ of the prime divisors W_j of $C^{(d)}$ passing through ζ, and $a_j = v_{W_j}(F)$. Applying v_{η_0} and using Claim 6:

$$v_{\eta_0}(F) = \sum_j v_{W_j}(F) \cdot \operatorname{ord}_\zeta(w_j).$$

By Claim 5, all terms with $W_j \notin \{\Delta_{P_1}, \ldots, \Delta_{P_s}\}$ are divisible by e; by Claim 7, $\operatorname{ord}_\zeta(w_i) = 1$ for the boundary divisors. Therefore, using Claim 8,

$$v_{\eta_0}(F) \equiv \sum_{i=1}^{s} c_i \equiv 0 \pmod{e},$$

which is (1). As explained above, this shows $\chi(\rho'(I)) = \{1\}$ for every character χ of G', hence $\rho'(I) = \{1\}$; combined with $\rho_p(I) = \{1\}$ this gives $\rho(I) = \{1\}$, i.e. $\mathcal{Q}$ is unramified at (H, η_0). $\square$

Conclusion of the proof

Proof of Lemma 1. Let $G = \Lambda^\times$ and let $\mathcal{P}^{[d]}$ be the G-torsor on $U^{(d)}_{\text{ét}}$ corresponding to $\mathcal{F}^{[d]}$ under Proposition 9, with character $\rho : \pi_1^{et}(U^{(d)}, \bar{u}) \to G$. By Lemma 2(3), the only codimension-one point of $\tilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$ is η_0, so the hypothesis that $\mathcal{F}^{[d]}$ is tamely ramified along H means precisely that $\mathcal{P}^{[d]}$ is tamely ramified over $\tilde{C}^{(d)}_{\mathfrak{m}}$ at (H, η_0) (Definition 54). By Proposition 4, $\mathcal{P}^{[d]}$ is then unramified at (H, η_0). By Proposition 3, the character ρ factors uniquely as $\rho = \tilde{\rho} \circ \iota_*$ through $\pi_1^{et}(\tilde{C}^{(d)}_{\mathfrak{m}}, \bar{u})$. Let $\tilde{\mathcal{F}}^{[d]}$ be the locally constant sheaf of free rank-one Λ-modules on $(\tilde{C}^{(d)}_{\mathfrak{m}})_{\text{ét}}$ corresponding to $\tilde{\rho}$ (Corollary 14 together with Proposition 9). Since $\tilde{\rho} \circ \iota_* = \rho$, its restriction to $U^{(d)}$ is isomorphic to $\mathcal{F}^{[d]}$. $\square$

Remark. The proof also gives uniqueness: since $\iota_* : \pi_1^{et}(U^{(d)}) \to \pi_1^{et}(\tilde{C}^{(d)}_{\mathfrak{m}})$ is surjective (Proposition 3), the extension $\tilde{\mathcal{F}}^{[d]}$ is unique up to isomorphism, and more generally the restriction functor from locally constant sheaves on $\tilde{C}^{(d)}_{\mathfrak{m}}$ to locally constant sheaves on $U^{(d)}$ is fully faithful. This is the source of the uniqueness assertions in Theorem 87 and Theorem 2.

Remark. Statements of this kind appear in both of the works our approach follows, with different proofs. Guignard ([Gui19], Lemmas 5.7 and 5.9) does not extend $\mathcal{F}^{[d]}$ over a compactification: he shows that $\mathcal{F}^{[d]}$ is constant on the (affine space) fibers of Φ_d, using the triviality of the tame fundamental group of $\mathbb{A}^1$, and then descends it along Φ_d directly by a duality argument. Takeuchi ([Tak19], proof of Theorem 1.2 in §4) obtains the extension of the corresponding character over $\tilde{C}^{(d)}_{\mathfrak{m}}$

from his refined (Kato–Matsuda) Swan conductor computation ([Tak19], Theorem 2.7 and Corollary 2.8), which for p-power order characters shows unramifiedness at η_0 directly. The proof given here is instead elementary: beyond the purity of the branch locus, it only uses Kummer theory and the geometry of the linear systems $|\mathcal{L}|$ inside $C^{(d)}$.

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